



**DGLRM**

# **From Bed Rest to Mars:**

## **Development of a Literature-Derived Machine Learning Framework for Predicting Lower-Limb Muscle Atrophy in Spaceflight Analogues**

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**64. Jahrestagung der DGLRM**

# Agenda

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Which muscles lose most

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Findings, limits, next steps

*The marker in the bottom-right corner of each slide shows where we are.*



# Motivation: can we predict muscle atrophy?

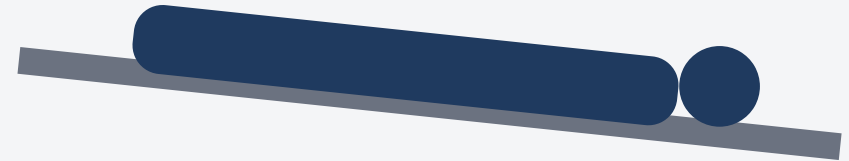
## What are we doing?

- Lower-limb muscle atrophy is a major challenge in long-duration spaceflight
- Research question: can the extent of muscle atrophy be predicted?
- Aim: a literature-derived dataset and a machine-learning framework for this prediction

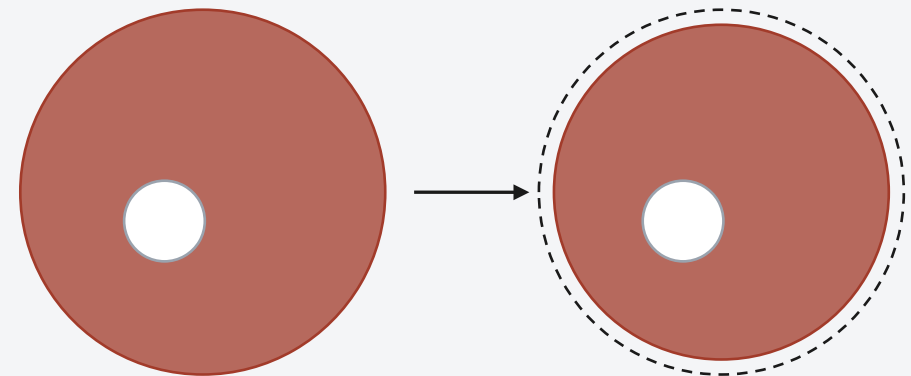
## Why is this important?

- Prevention: identify risk early and leave time to intervene
- Countermeasures: match exercise and nutrition to the predicted risk
- Vulnerable muscles: show which muscles need closer monitoring
- Function: less muscle loss helps preserve strength and power
- Mission planning: estimate muscle loss on long missions with limited exercise equipment

### Ground analogue: 6° head-down bed rest



*Weeks to months in bed; legs carry no body weight*



**Before bed rest**

**Day 60: -16%**

*Calf cross-section to scale; dashed line = size before. Model estimate for control groups (plantar flexors, -15.8% at day 60).*

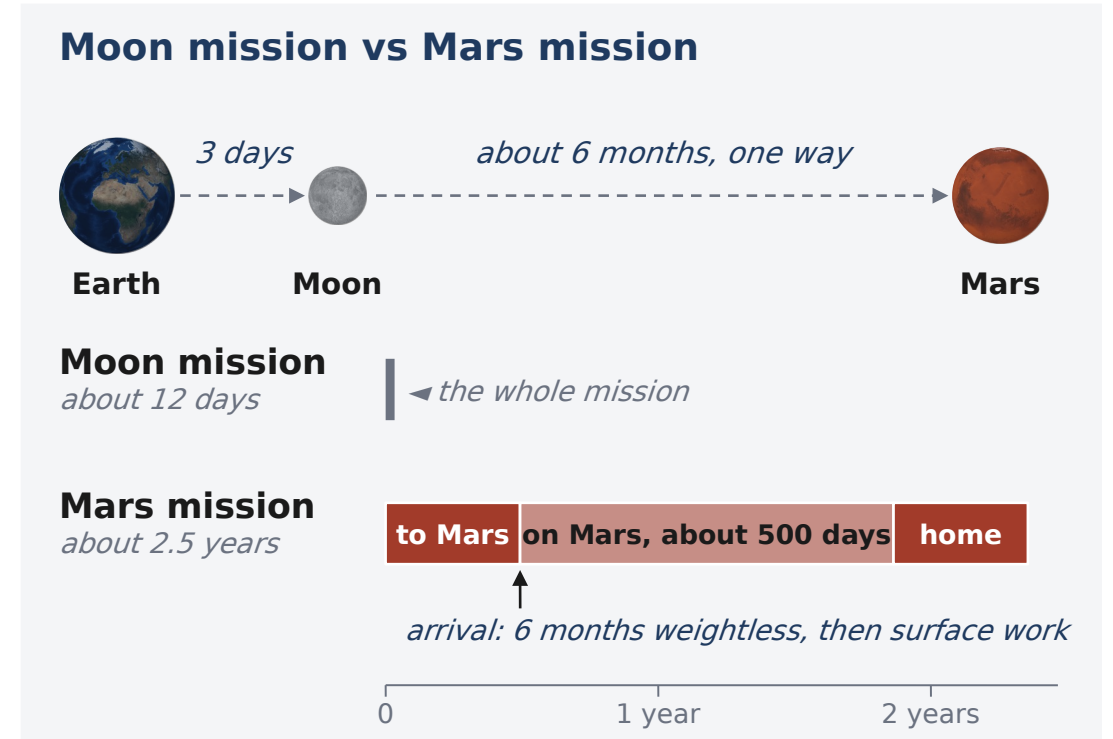
# Why from bed rest to Mars?

## Bed rest: the ground analogue

- Direct spaceflight data on muscle are scarce
- Head-down bed rest is the established ground analogue: it unloads the legs under controlled conditions
- Not equivalent to spaceflight, but a sound basis to build and test prediction models

## Mars: the long-term application

- A Mars mission lasts years, not days: about 6 months of transit each way
- No early return in an emergency: the crew depends on its own fitness
- After months in weightlessness, the crew must leave the spacecraft and work on the surface



# What range did we study?

**Abstract**  
15 bed-rest studies

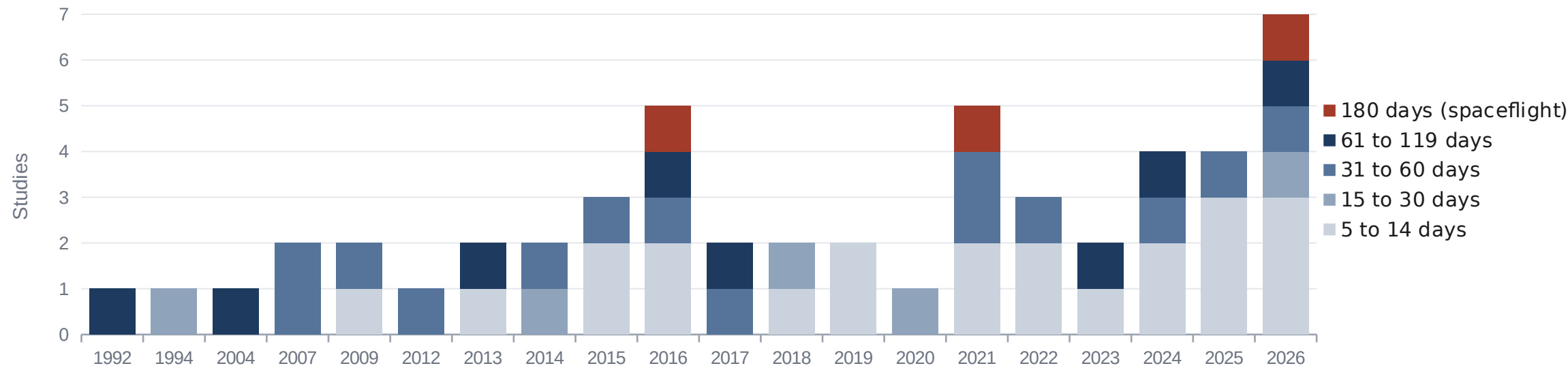


**Final dataset**  
52 studies

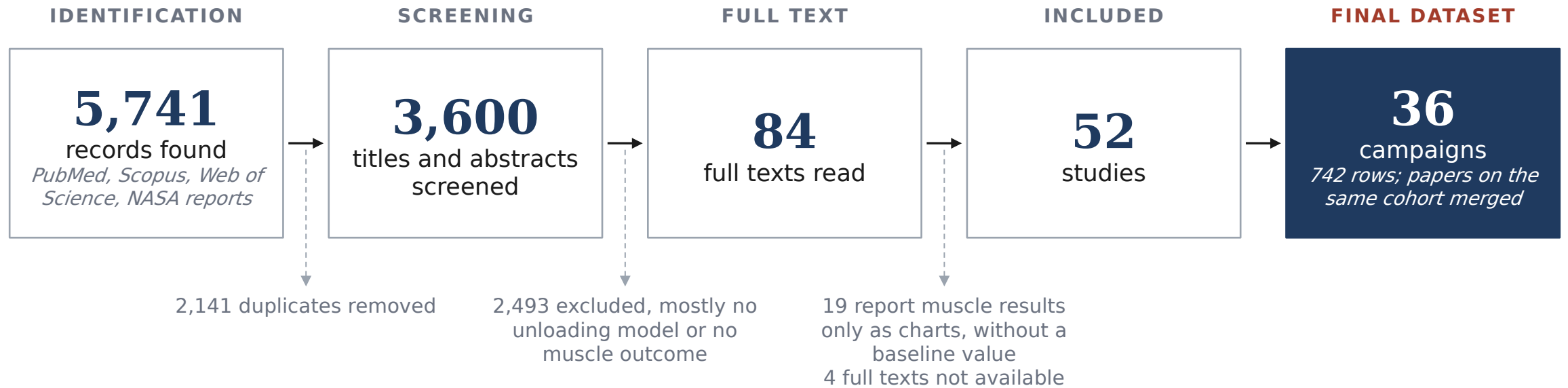


Days of unloading 0 30 60 90 119 150 180

Studies published per year, by length of unloading (years without studies omitted)



# How did we build the dataset?



*Flow of records in the style of PRISMA 2020, the standard for reporting systematic reviews.*

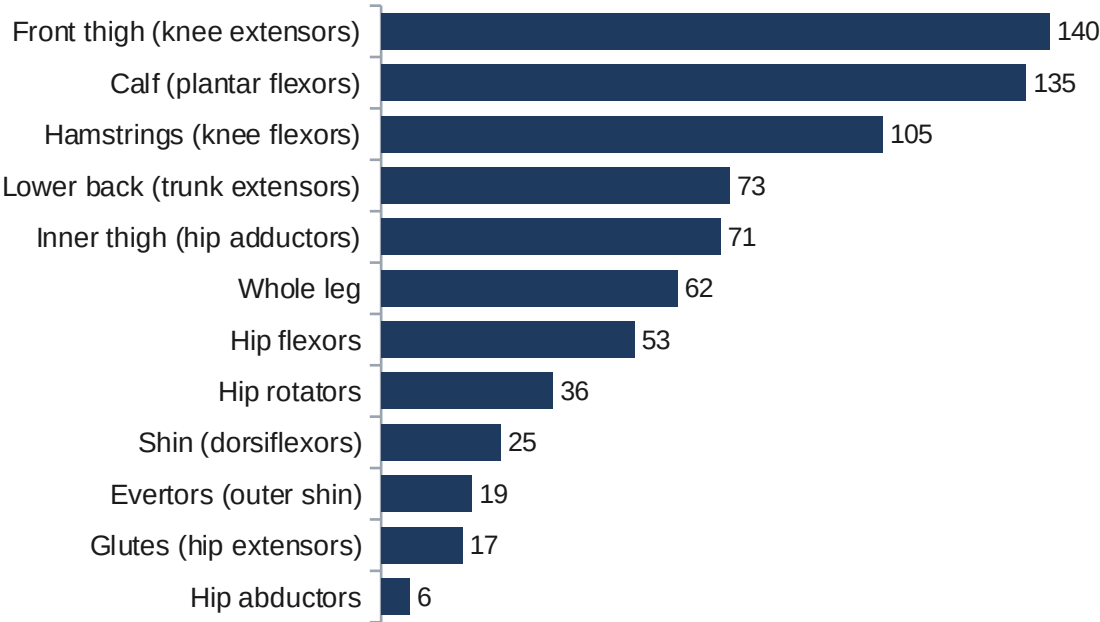
# What does the dataset look like?

One row = one muscle outcome of one group at one scan day

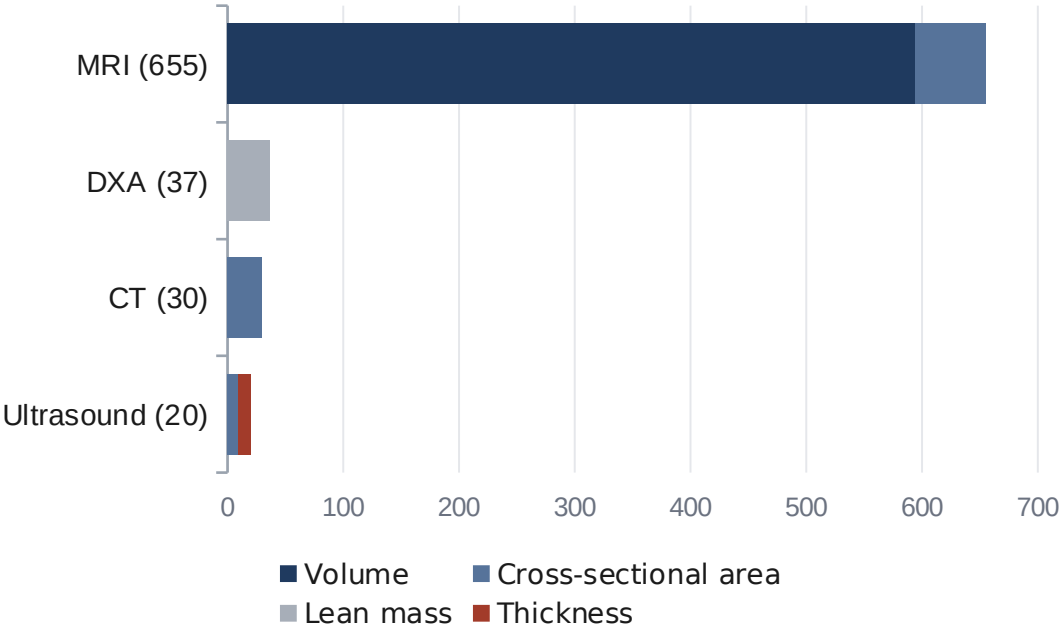
| Unloading | Scan day | Muscle          | Participants         | Countermeasure | Method | Size measure | Target: % change |
|-----------|----------|-----------------|----------------------|----------------|--------|--------------|------------------|
| 60 days   | 59       | gluteus maximus | 8; 25% women; age 34 | none (control) | MRI    | volume       | → -9.4%          |

Example row from the AGBRESA bed-rest campaign (Tran et al., 2021). Target: percentage change in muscle size from before unloading.

742 rows across 51 muscles, by muscle group



Rows by imaging method and size measure



# Are all muscles affected the same?

## **PLACEHOLDER: content by Niloufar**

*Suggested points from the final report:*

- No. Calf muscles (plantar flexors) lose most: –15.8% at day 60, 5.7 pp more than the front thigh
- Front thigh, back thigh and shin muscles lose around 10%; hip muscles least
- 30% of the differences between measurements lie between muscles within the same study
- Groups with countermeasures lose 3.7 pp less than controls

*Possible figure: muscle ranking (figures/F3\_muscles)*

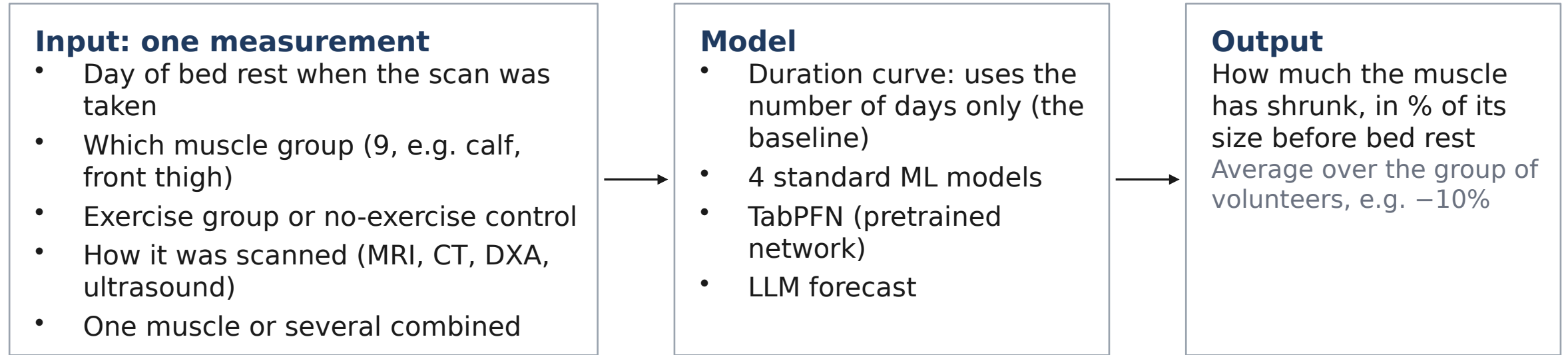
# From data to forecasts

## **PLACEHOLDER: content by Niloufar**

*Suggested points from the final report:*

- Bridge to part 2: the data show that loss depends on time and on the muscle
- Next question: can a model forecast the loss in a study it has never seen?

# Every model is tested on a study it has not seen



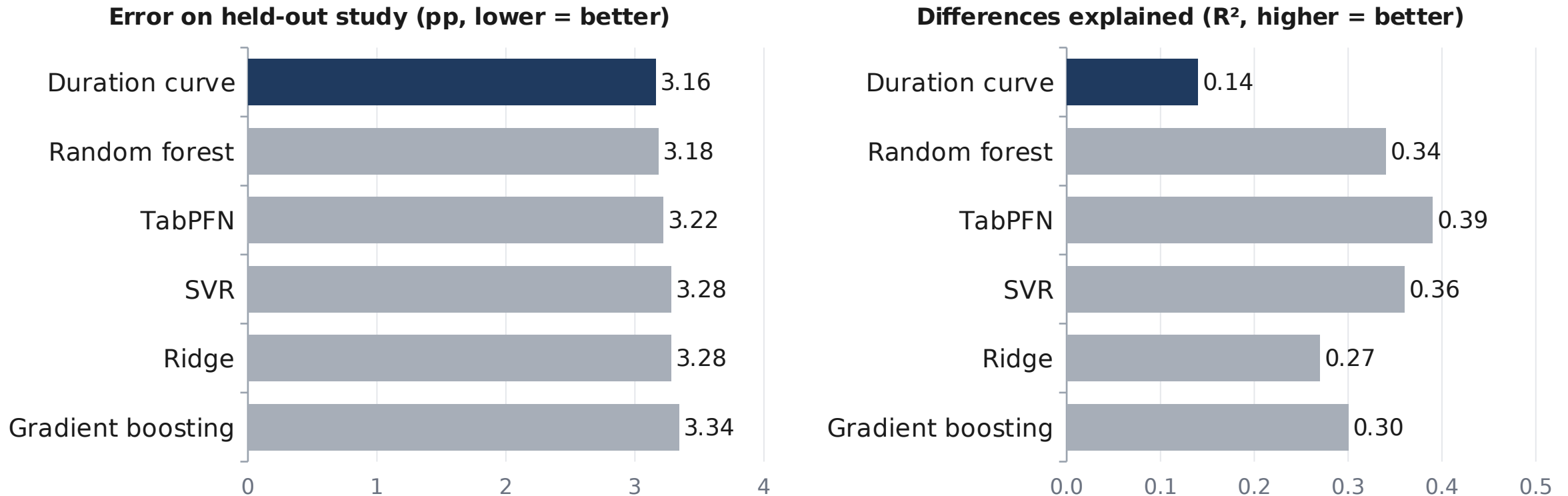
## Testing: hold out one study, train on the rest, repeat 32 times



Grey: 31 training studies. Red: the held-out test study. Each study is held out once.

- Why studies, not measurements: the 346 measurements come from only 32 independent studies, because several papers report the same volunteers.
- Score: average error in percentage points (pp). Forecast -10%, measured -13%: error 3 pp. Each study counts once.

# No ML model has lower error than the duration curve



- $R^2$  is the share of the differences between measurements that a model explains: 0 is none, 1 is all. The ML models explain more (0.27 to 0.39 vs 0.14) because they learn which muscles shrink more.
- Their error on a new study is still no lower. Most of the error left is a shift that affects a whole study (its scanner, volunteers, protocol), and none of our inputs describe it.

# LLM forecast: what is sent and what comes back

## Request (one per measurement)

### participants

men only, mean age 35, n = 11

### protocol

head-down tilt bed rest, 21 days, control group

### measurement

vastus lateralis, MRI cross-sectional area

### target

day 21 of bed rest

### typical\_curve\_other\_campaigns

duration curve fitted on the other 31 studies

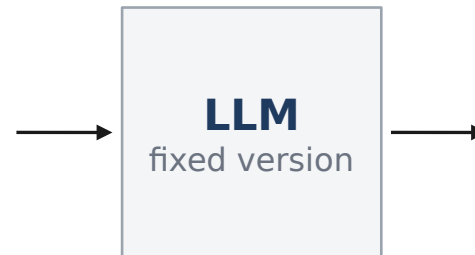
### observations\_other\_campaigns

rows from the other 31 studies, most similar first, up to about 26,000 tokens

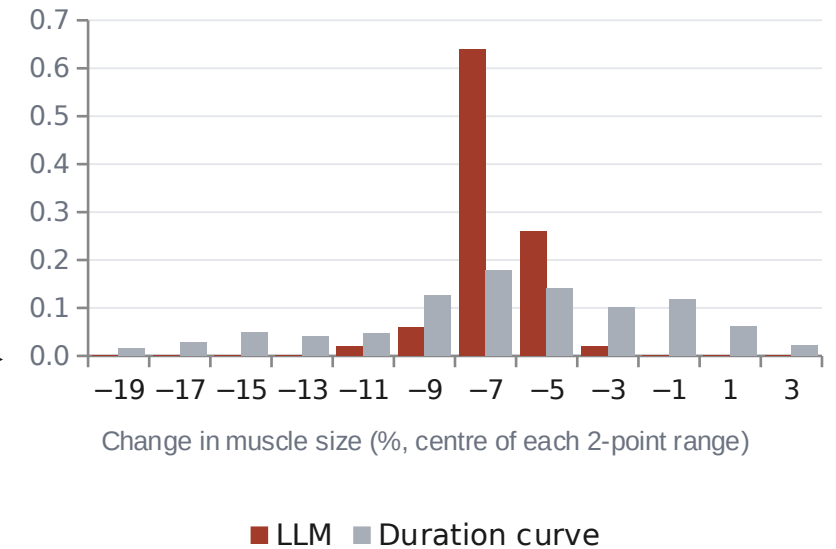
### question

Which range will the change fall in? 19 options: below -30%, 2-point ranges up to +4%, +4% and above

*Field names as sent; "campaign" means study.*



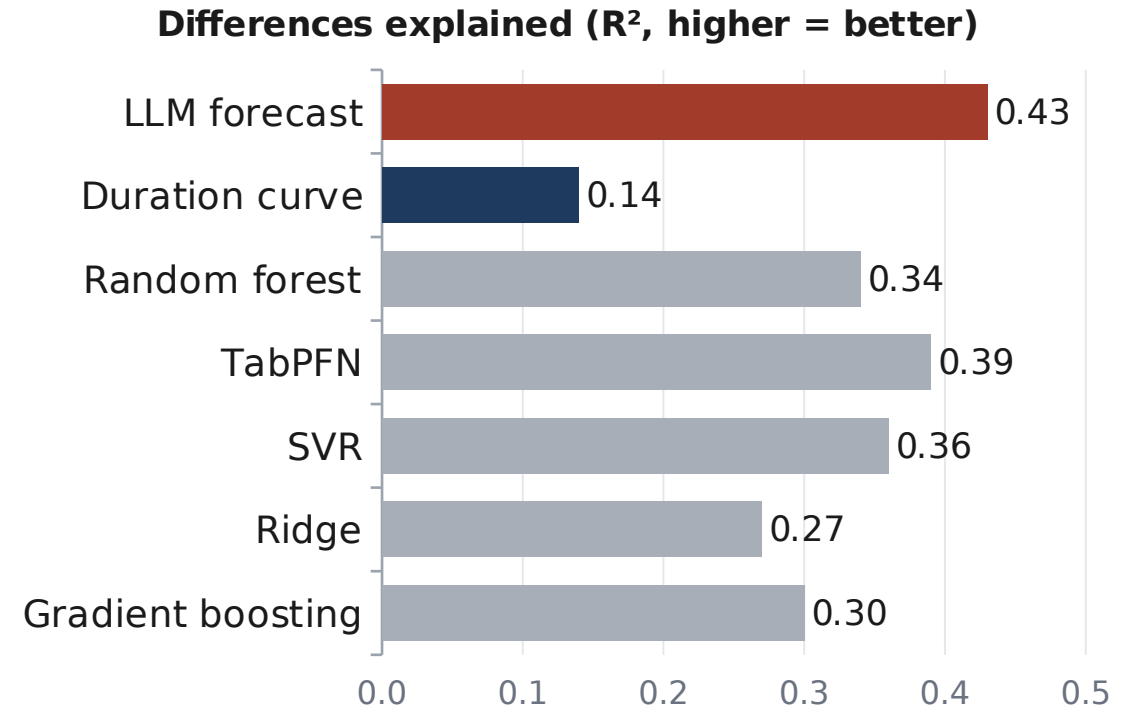
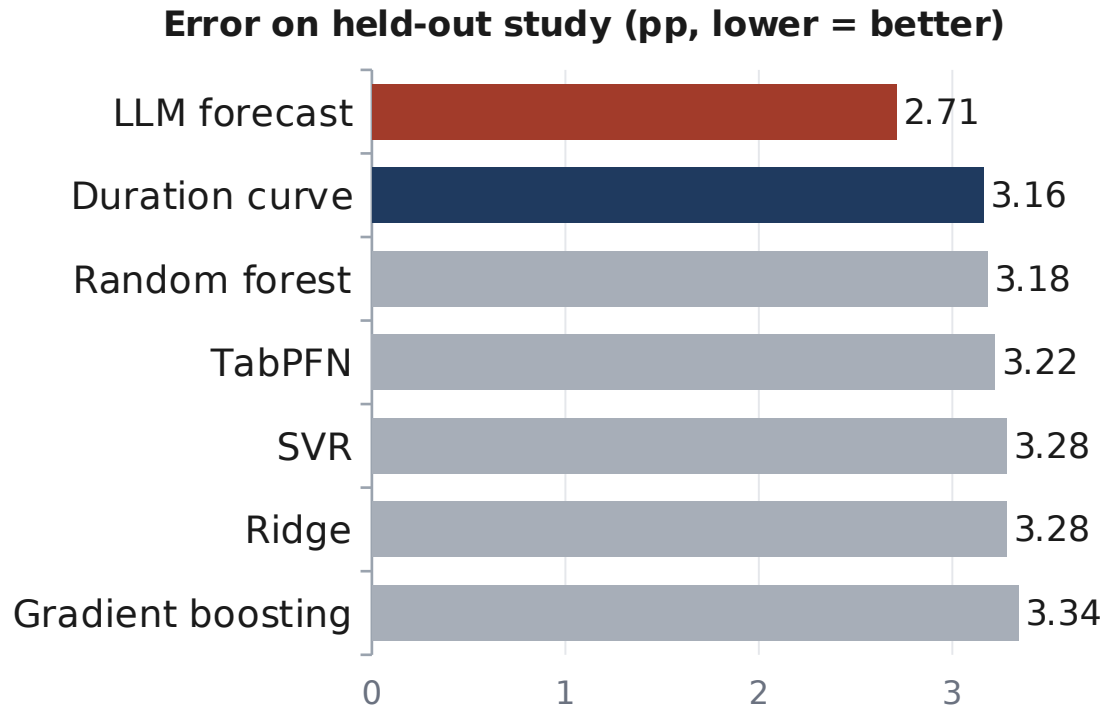
## Answer: probability per range



- Our code turns the probabilities into one forecast, their weighted average: -6.6%
- Measured: -9.4%, an error of 2.8 pp. This is a typical row: half of the LLM's errors are smaller.

**Never sent (checked on every request):** the held-out study's measurements, and any paper, author or study name. All 430 answers are saved, so every result can be rebuilt without calling the model again.

# The LLM forecast has the lowest error of all models



- Same measurements, scored the same way: error 2.71 vs 3.13 pp, so 0.42 pp smaller. Closer than the curve in 20 of 32 studies.
- 95% confidence interval of that difference: 0.13 to 0.73 pp. The whole range is above zero, so it is unlikely to be luck. No other model achieves this.

*The curve shows 3.13 pp here, not 3.16, because the comparison scores it exactly as it scores the LLM.*

# Validity checks on the LLM forecast

*Gain = how much smaller the LLM's error is than the curve's, in pp (full result: 0.42). Each check and its pass rule were fixed before running it.*

## Does it recognise published studies?

- Everything identifying a study removed: gain still 0.30 pp, interval above zero
- Asked to choose the study's name from a list of options: right 19% of the time, where guessing gives 8%

## Does it use the other studies' data?

- Their values shuffled: worse than the curve (gain  $-1.25$  pp)
- None of their data given: error 5.85 vs 3.13 pp

## Is it sensitive to presentation?

- Other studies' rows in a different order: error changes 3%
- Answer ranges moved by 1 point: error changes 1%

## Is it reproducible?

- 20 identical requests resent: forecasts differ by 0.17 pp on average, well below the 0.42 pp gain

## Limitations of this result

- Added after the ML result was known, so it was not part of the original plan
- Overconfident: when it is 80% sure the value is in a range, it is there only 62% of the time. We therefore report only its single-number forecast.
- When also given a study's own earlier scans, it does not make use of them

# Conclusions, limitations and further work

## Conclusions

- With 32 independent studies, no ML model forecasts better than a curve based on days alone (best 3.18 vs 3.16 pp error).
- An LLM that reads each study's description and the other studies' data cuts the error to 2.71 pp (curve 3.13 pp), and this holds under every check.

## Limitations

- Only 32 independent studies
- No data beyond day 119
- Mostly young men (347 of 425 measurements)
- Group averages, not individual people
- Bed rest imitates spaceflight but is not the same

## Further work

- Fix the LLM's overconfident ranges
- Check results without the largest study, and with MRI scans only
- Extend the forecast to a 180-day mission, with an uncertainty range

# Backup: is the dataset good enough?

## Did you miss studies?

- Searched PubMed, Scopus, Web of Science and NASA reports from 2013 on (5,731 records), plus 9 older studies
- Not covered: Embase (no access) and 1,023 records not screened in time

## One study dominates the data

- MEDES 90-day: 40% of all rows, a quarter of the modelling data
- Scoring gives each study one vote; the re-run without it is still to do

## Can MRI, CT, DXA and ultrasound be mixed?

- Each method gets its own term: DXA and ultrasound thickness show 5 to 6 pp less loss than CT; MRI volume is within 0.5 pp
- The MRI-only re-run is still to do

## How reliable is the extraction?

- Every value is traced to its paper and page, table or figure
- 674 of 742 rows high confidence; 19 read off figures; no independent second extraction

# Backup: is the LLM gain real, and does it matter?

## Is 0.42 pp worth anything?

- It is 13% lower error than the curve (3.13 to 2.71 pp)
- Small: we report it as a finding, not the headline

## 20 of 32 studies is close to a coin flip

- Counting wins only:  $p = 0.22$  (sign test)\*
- The size of the wins matters: mean gain 0.42 pp, 95% CI 0.13 to 0.73

## Does it rest on a few studies?

- 3 long MRI studies (NASA SPRINT, Berlin BBR2-2, MEDES LTBR) give 56% of the gain; without them 0.20 pp, CI  $-0.01$  to  $0.43^*$
- Only SPRINT is recognised, and with study details removed the LLM still wins there (3.79 vs 5.92 pp)

## Did it read these papers in training? Was it added after the fact?

- Training exposure cannot be ruled out by any test; hiding study details and the naming test are the strongest checks available
- Added after the ML result, but every check and target was committed before the first answer

*\* Our calculation from results/forecast\_predictions.csv; not in the report.*

# Backup: why did ML fail, and was the comparison fair?

## Did the ML models get a fair chance?

- Tuned inside each training fold, grouped by study; TabPFN needs no tuning
- All models saw the same inputs, folds and scoring

## Why not add age, sex or countermeasure type?

- 32 studies leave room for about 3 study-level inputs (about 10 studies per input)
- Age and sex are fixed within a study (347 of 425 rows men-only), so they act as a study label; 126 countermeasure rows spread over 9 types

## Is the duration curve a straw man?

- Ridge regression uses days, muscle and every other input: 3.28 vs 3.16 pp
- Knowing the muscle raises  $R^2$  but not the error on a new study

## Would a nearest-neighbour method match the LLM?

- Not tested. If the gain comes from picking similar rows, this is the obvious next control

# Backup: can this be used to plan a Mars mission?

## A Mars transit is about 180 days; the data stop at 119

- Only one study reaches 119 days
- A 180-day estimate is planned from the duration curve only, with an uncertainty range, labelled as an assumption

## Which countermeasure works? What about individuals?

- Countermeasure groups lose 3.7 pp less (95% CI 1.5 to 5.9), all types pooled
- Every row is a group average; nothing about individual risk

## Bed rest is not spaceflight

- Only 14 spaceflight rows, 8 of them one back muscle: too few to test transfer

## Can anyone reproduce an LLM result?

- Fixed model version; all 430 answers saved, so every number rebuilds without the model
- Identical requests differ by 0.17 pp on average